- The location of the ribosomes in a cell is a determining factor of the type of protein produced. If the ribosomes are free floating throughout the cell, the proteins that are used within the cell are produced. When ribosomes are attached to endoplasmic reticulum (referred as rough endoplasmic reticulum or rough ER), the proteins that are used inside the cell or outside the cell are produced.
- The catalytic activity of the ribosome is carried out by the RNA.

1.4.8.Lysosomes

These are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus. The isolated lysosomal vesicles have been found to be very rich in hydrolytic enzymes, called hydrolases such as lipases, proteases, carbohydrases, which are optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

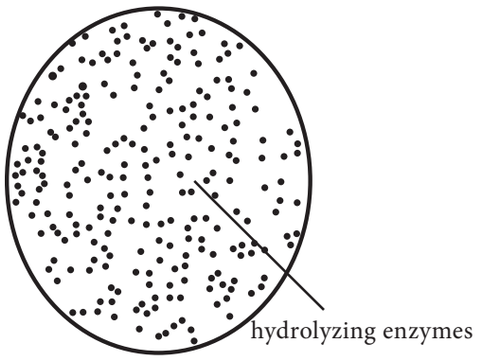

Figure 1.11 A schematic diagram of lysosome

1.4.9.Peroxisome:

Peroxisomes are microbodies that are abundantly present in mammalian liver and kidney, and also in plant cells. It depends on the type of eukaryotic cell. The matrix of Peroxisomes is rich in enzymes but a few enzymes are located in the membrane. The common enzymes present in the matrix of peroxisomes are catalases and peroxidases which metabolize a number of substrates. Enzymes present in the membrane of peroxisomes are cytochrome b5 and NADH cytochrome b5 reductase.

Functions of peroxisomes

- A major function of the peroxisome, in yeast and plant cells are to breakdown the fatty acid molecules, in a process called beta-oxidation. Peroxisomes are involved in lipid biosynthesis
- Peroxisomes contain enzymes required for the synthesis of plasmalogens
- Peroxisomes in seeds are responsible for the conversion of stored fatty acids to carbohydrates, which is critical in providing energy and raw materials for growth of the germinating plant.

1.4.10.Cytoplasm:

The ground substance that fills the interior of the cell is called cytosol or cytoplasm. It is a jelly-like substance and it is made up of eighty percent

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